

ZIHUI LU

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TECHNICAL SKILLS

Languages: C/C++(STL, Valgrind), C#/.NET, Python(PyTorch, NumPy), Javascript/TypeScript, SQL(Postgres), HTML/CSS

Databases: SQL, PostgreSQL, Microsoft SQL Server, Supabase, Cloudbase

Frameworks: React, Next.js, Node.js, .NET Core, Octokit

Tools: Git, Jenkins (CI/CD), Jira, Docker, EZGL, MATLAB

Systems: Linux/Windows Development, REST APIs, Algorithms Optimization

PROFESSIONAL EXPERIENCE

SOTI Inc. | *Software Developer Intern*

May 2022 – Aug 2023

- Designed a **C#** tool using **Octokit SDK** to auto-generate PRs for managed components, reducing manual steps by **90%** and saving **3+ hours per task**.
- Integrated tool into Jenkins pipeline, improving team deployment frequency.
- Built a **C#** service to track **500+** BDD tests in real-time, reducing failure diagnosis time from **2 hours to 30 minutes**.
- Investigated and resolved BDD test failures to maintain testing process reliability.

Shenzhen LaitCloud Technology | *Software Developer Intern & Media Content Creator*

Jan 2026 – June 2026

- Built and deployed a production multi-tenant **SaaS** for construction finance tracking using **React/TypeScript** (frontend) and **Node.js/MongoDB** (backend), serving real users with role-based access and **JWT** authentication. [↗ Live Demo](#)
- Produced promotional video content using **CapCut Pro**, AI agents, **Remotion**, and **Hyperframe**.

EDUCATION

University of Toronto

Graduated May 2025

B.Eng in Computer Engineering (Minor: Artificial Intelligence & Business)

Key courses: Data Structures & Algorithms, Computer Networks, Databases, Machine Learning

TECHNICAL PROJECTS

Toronto GIS Map Engine (*C++, EZGL, Valgrind*)

Spring 2021

- Developed a **high-performance** map renderer using **Dijkstra's algorithm** (95% query success rate).
- Reduced memory leaks by **70%** through Valgrind profiling and smart pointer refactoring.

Face Mask Detection System (*Python, PyTorch, CNN*)

Fall 2021 - Winter 2021

- Trained a **CNN model** with **96% accuracy** (vs initial 65%) by tuning hyperparameters (batch size, epochs) and applying **k-fold cross-validation**.
- Processed 10,000+ images using NumPy and PyTorch and implemented data augmentation to reduce overfitting.

School Food Program Dashboard (U of T) [↗ Live Demo](#) (*Next.js, TypeScript, React*)

Fall 2024 - Spring 2025

- Led a team to optimize frontend performance by **78%** (PageSpeed 49 → 87) using **code-splitting** and **SSR**.
- Implemented REST API endpoints in Node.js to serve nutrition data to **1,000+** monthly users.
- Developed interactive data visualizations (DataWrapper & Flourish) to highlight regional nutrition trends.

ADDITIONAL

- Languages:** English, Mandarin, Cantonese
- Leadership:** Tutored 30+ high school students in STEM subjects, improving grades by 20% on average.
- Technical Writing:** Maintain detailed project documentation